

Payments Guidelines Query Expert

Answers to payment questions, with sources you can check

[Team ID] | [Team member names]

Problem Statement and Motivation

The Why / For product, operations and compliance teams in banks and fintechs

01

Find the requirement

One question may span several circulars and later clarifications.

02

Understand conditions

A relevant passage can still omit a qualification elsewhere in the source.

03

Verify the answer

Users need to see which guideline supports each important statement.

Goal: reduce research effort while keeping source checking part of the task.

Proposed Solution

A plain-language question, a cited answer and a source the user can inspect

QUESTION ASKED

Can I use UPI Tap & Pay at a shop's card machine?

Draft answer — check the supporting excerpts.

UPI Tap & Pay can be used at a shop's Point-of-Sale (PoS) terminal if the terminal is NFC-enabled and certified in accordance with PCI PTS / CPOC standards. **S1** The facility is restricted to Person-to-Merchant (P2M) transactions only. **S4** Payee PSPs must integrate the Point-of-Sale Common Library for UPI at the PoS terminal. **S10**

Automated support checks passed for the displayed claims. Review details below.

Follow S1

UPI OC 186A
Page 1

NFC means
contactless
tap technology.

Also available: guideline search and inspection, summaries and document comparisons.

Data Sources and Tools

The How / An attributed knowledge base, starting with NPCI

From circulars to search indexes

- 1 Split by page, heading and clause
- 2 Long blocks: ≤ 220 words; 35 overlap
- 3 Keep document ID, page and links
- 4 Index text in BM25; vectors in Chroma

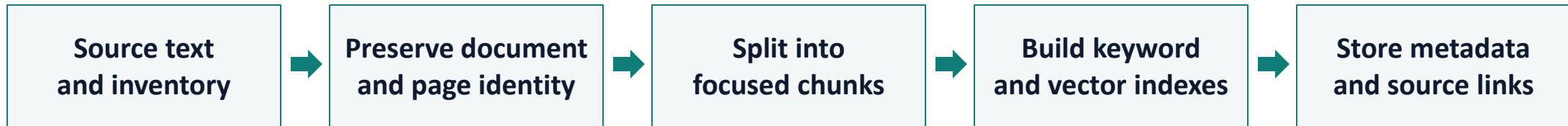
Local model and storage stack

Qwen3.5 9B / answer generation
Qwen3 embeddings / semantic search
MiniLM / candidate reranking
Chroma + SQLite / vectors + metadata

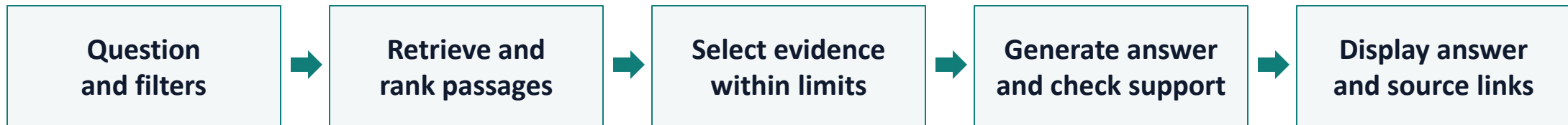
How the System Works

The How / Retrieval brings source evidence into the model's context

Prepare



Answer



Exact search finds references; semantic search finds similar meanings.

What the Preliminary Evaluation Tells Us

Purpose: does more complex search find better evidence? Same 50 UPI cases and corpus.

Search method	Evidence found	Negatives with no result	Search time (ms)
BM25	55/55	6/6	12.0
Vector	55/55	5/6	83.0
Hybrid	55/55	5/6	78.0
Hybrid + reranker	55/55	5/6	160.5

55/55 = all expected passages found. 6/6 = no passages returned for all negative cases.

Implementation Roadmap

From a working prototype to an evaluated, repeatable local demonstration

NOW · Cited Q&A, source reader, summaries and comparisons implemented

Weeks 1–2

Validate answers

Review the 50 cases.
Freeze unseen questions.

DELIVERABLE

Reviewed test set +
answer-quality scorecard

Weeks 3–5

Improve reliability

Fix source and answer failures.
Shorten long summaries.
Compare search settings.

DELIVERABLE

Tested revision +
quality / latency comparison

Weeks 6–8

Test with users

Observe research tasks.
Package setup and final demo.

DELIVERABLE

User-task results +
repeatable demonstration

Indicative weeks after midterm; reviewer access is a dependency. Another payments source is an optional later extension.

Discussion and Feedback

Feedback requested on the next stage of this capstone

01 Scope

Is the question-to-source workflow useful and manageable?

02 Evaluation

Which multi-document and missing-evidence cases should come first?

03 Usefulness

What task outcomes would demonstrate value to payments teams?

Immediate priority: validate answers and improve failures before expanding coverage.